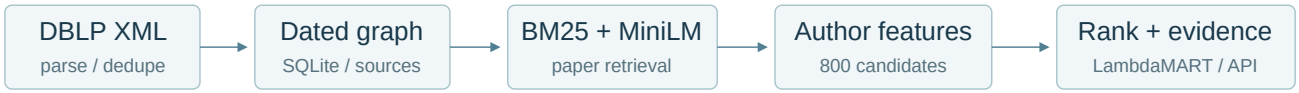
TECHNICAL ASSIGNMENT · DESIGN DOCUMENT · 6 NOVEMBER 2026

ScholarGraph

Evidence-grounded expert discovery from DBLP. A local-first reference implementation, not a hosted production service.



Offline preparation precedes query-time retrieval and inference; diagram is not live telemetry.

1. Data and context graph

DBLP XML is streamed into SQLite. Stable source keys identify papers and author observations; repeat ingestion is idempotent. Historical snapshots exclude unknown-year and post-cutoff records. Nodes represent people, papers and venues. Typed author–paper, author–venue and coauthorship edges include publication date and source record. Search indexes and session state are separate from the read-only graph.

Resolution blocks plausible name variants, then uses names plus coauthors, venues and years to match noisy observations. The conservative synthetic evaluation reported precision 98.25% and known-author recall 16.76%; this is not a production merge rate. An incorrect merge is costlier because it combines different people's evidence; abstention can cause a split and missed discovery.

2. Retrieval and model

BM25 and local MiniLM embeddings independently retrieve up to 1,024 historical papers per channel. The system aggregates paper matches into at most 800 author candidates and applies an 80-tree person-level LambdaMART ranker trained in this project. Features include lexical and dense relevance, title overlap, publication counts, temporal signals and collaborator count. The model is selected and frozen before its historical test. Scores are ordering signals, not probabilities.

> The 500-query temporal test is now inspected: nDCG@10 0.0354 and MRR 0.0488 within up to 800 candidates. Future authorship is incomplete proxy supervision, not a human relevance or hiring-accuracy label.

3. Training, evaluation and limitations

The fixed 165,464-paper source spans selected DBLP venues and 1995–2025. The training graph ends in 2020; 493 later paper titles provide queries, with 270 groups containing retrieved positives and contrasts. Future authors receive positive proxy labels; non-authors used as contrasts are unjudged. Validation uses a 2023 graph and 129 later queries. A 2024 graph and 500 reserved 2025 titles form the historical test. Source IDs, dates and exact normalized titles are checked for leakage; semantic near-duplicates and identity errors remain possible.

On the 500-query test, lexical / dense / hybrid nDCG@10 were 0.0245 / 0.0268 / 0.0313 versus 0.0354 for the trained ranker. A retrained ablation removing explicit recency features scored 0.0334; candidates and other time-related signals were unchanged. MRR is over each returned list up to 800, not MRR@10. These low proxy results and the missing human relevance judgments limit conclusions. The live search's lexical-overlap abstention can reject relevant synonyms; source selection can omit older relevant work.

4. Assistant, grounding and failure handling

The graph-tool backend supports schema-validated search, profile and collaboration-path operations, with shortlist/reject decisions and deletable session memory. Freshly prepared backends use a schema-validated local LLM for query understanding, with deterministic fallback when unavailable or invalid. Typed explanation claims are checked against authorship, dated edges and source records; the verifier does not prove arbitrary generated prose true. Ten historical scripted scenarios passed; 0 of 191 checked typed claims were unsupported and 8 of 8 scripted query attacks were blocked. These are bounded tests, not general guarantees.

The search page exposes the graph assistant and shared decision memory beside results. Graph operations use explicit commands. With Gemini configured, unsupported questions can fall back to general chat without graph data or tool access; answers are marked unverified. The separate conversation workspace also retains Gemini/local chat. Generated prose is not covered by the typed verifier.

5. Alternatives and trade-offs

SQLite and in-process BM25/vector search keep installation, offline execution and deployment simple for the assignment slice. PostgreSQL with durable workers, Redis and a vector/search service would support concurrent production workloads at the expense of operations and network dependencies. MiniLM is small and local; larger encoders or cross-encoders cost more CPU and did not show a robust enough measured improvement here. A GNN or graph expansion could use context more deeply but would add complexity and require unbiased relevance evaluation.

6. Privacy and trust

DBLP is public but derived research profiles and session decisions still merit care. Search graphs are read-only; mutable decisions and conversations live in separate local SQLite state. Do not ship raw corpora, session data, API keys or local model caches. Remote providers receive submitted prompts and any selected research/document context. The current local application has no authentication or tenant isolation and must not be exposed as a multi-user service.

7. Path to 50 million papers and 200 queries/s

This is a proposed design, **not a demonstrated capacity claim**. Partition ingestion by source and time; use resumable, idempotent jobs with source versions and validation. Store canonical entities and provenance in a partitioned relational store; precompute embeddings and person-level aggregates asynchronously. Keep immutable versioned snapshots for temporal evaluation. Stream updates to a search/indexing layer and publish graph/index versions together.

Shard a BM25 index and an approximate-nearest-neighbor vector index by corpus partition, with replicas across availability zones. Query both concurrently under explicit timeouts; cap paper and author fan-out; cache popular queries and normalized features; rerank a bounded candidate set; fetch explanations in batches. Partition high-degree graph neighborhoods and keep provenance lookups separate from broad traversal. A queue, backpressure, graceful lexical-only fallback and request-level traces prevent one slow component from exhausting capacity.

Use load tests and service-level telemetry to budget p95: for example network/auth 30 ms, parallel retrieval 150 ms, aggregation/reranking 100 ms, evidence lookup 100 ms and remaining overhead 120 ms. These are illustrative allocations, not measurements. Measure ANN recall versus exact search, rebuild lag, replica memory, high-degree skew and failure modes. Two hundred queries/s at p95 500 ms depends on hardware, corpus growth and workload; this SQLite implementation has not demonstrated it.

8. Future product directions

Deeper graph: use versioned, dated, sourced multi-hop relationships and query-specific evidence with degree controls. **Outreach:** drafts cite selected evidence and require explicit human approval before any external send, with durable audit and idempotency. **Controllable memory:** show, edit, expire and delete saved decisions, isolate tenants, and provide retention policies. **Feedback:** capture explicit, consented judgments and review bias before using them for offline training; do not equate user actions with ground truth.

9. Operational and submission status

The batch command and 200-paper sample work offline with prepared dependencies. Compose uses a sample SQLite graph; image build needs dependency/model downloads. Compose startup has not been verified on this machine because the Docker Desktop daemon fails. The local chat model is prepared during the image build; independent human relevance review remains incomplete. Current local and CI results must be rechecked on a clean environment before submission. A video is intentionally omitted at the user's request although the assignment asks for one.

Supporting evidence: `resources/results/README.md`, `lean-final-v1.json`, `recovery-identity-v2.json`, `lean-assistant-v1.json`, and the project journal. See the repository README for setup and reproduction.